Anna Toidze

Master's Student in Computational Biology and Bioinformatics

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Education

ETH Zürich, Department of Biosystems Science and Engineering — *Zürich, Switzerland*

Msc in Computational Biology and Bioinformatics — *Sept 2023 - Jun 2026*

- **GPA** 5.68/6.0
- **Excellence Scholarship & Opportunity Programme**: top 3 % of incoming MSc students, ≈50,000€ award covering living and tuition costs
- **Courses**: evolutionary dynamics, Bayesian phylodynamics, infectious disease dynamics, computational systems biology, ML for healthcare
- **Thesis**: "Pangenome Evolution of *Escherichia coli* on the Commensal–Pathogen Spectrum" written in Evolutionary Epidemiology of Infectious Diseases group supervised by Prof. François Blanquart in Collège de France, CNRS, and Prof. Julie Marin in Université Paris Cité, Inserm, IAME

Ruhr University Bochum, Faculty of Chemistry and Biochemistry — *Bochum, Germany*

BSc in Biochemistry — *Oct 2019 - Sept 2022*

- **GPA** 1.0/1.0, top student out of 20 in 2022 cohort
- **Wilke Award**: for best Bachelor's degree in Biochemistry
- **Courses**: biochemistry, general, organic, physical and analytical chemistry
- **Thesis**: "Mutational robustness and Physical stability of Fluorescent Proteins" written in Evolutionary Genomics laboratory under the supervision of Prof. Dr. Fyodor Kondrashov at Institute of Science and Technology Austria (ISTA)

Bioinformatics Institute — *Saint Petersburg, Russia*

Bioinformatics for Biologists, Additional Education — *Sept 2021 - May 2022*

- **GPA** 5.0/5.0, top student out of 53 of the year
- **Courses**: Python, R, command-line, data analysis and AI, discrete maths, molecular phylogenetics
- **Thesis**: "Search for homologs of egg-cell specific genes, a study of their expression patterns and regulatory elements for the creation of effective constructs for genetic engineering" under the supervision of Prof. Dr. Maria Logacheva

Authorized School No. 6 — *Tbilisi, Georgia*

High School — *Sept 2016 - May 2019*

- **GPA** 10/10, graduated with distinction rewarded with gold medal
- **Exchange program**: in 11th grade 3-month long student exchange program at Kantonsschule Hohe Promenade in Zürich, Switzerland

Research Experience and Internships

Institut Pasteur — *Paris, France*

PhD in Microbial Evolutionary Genomics supervised by Prof. Eduardo Rocha and Prof. François Blanquart — *Oct 2026 - Oct 2029*

- Studying the co-evolution of mobile genetic elements and their interactions with their hosts

Collège de France, CIRB, CNRS — *Paris, France*

Internship in Laboratory of Prof. François Blanquart and Prof. Julie Marin — *Nov 2025 - Sep 2026*

- Studying the pangenome evolution and niche adaptation in major *E. coli* clones

Roche, Pharma Research and Early Development (pRED) — *Basel, Switzerland*

Internship in Ophthalmology department under Prof. Sarah Doyle, Dr. Ema Ozaki, and Said Aktas — *Apr 2025 - Sep 2025*

- scRNA analysis in a pre-clinical mouse model of subretinal fibrosis investigating the effects of aging, **publication in PNAS**

ETH Zürich, Department of Environmental Systems Sciences — *Zürich, Switzerland*

Internship in Laboratory of Prof. Dr. Sebastian Bonhoeffer — *Nov 2024 - Mar 2025*

- Dynamically modelled the spread of mobilizable and conjugative plasmids carrying AMR genes and TA systems

University of Zurich, Department of Evolutionary Biology and Environmental Studies — *Zürich, Switzerland*

Internship in Laboratory of Prof. Dr. Andreas Wagner and Dr. Aleks Nivina — *Sept 2024 - Oct 2024*

- Stochastically modelled the recombination of antibiotic-producing polyketide system in *Streptomyces*

Okinawa Institute of Science and Technology — *Onna, Japan*

Laboratory Technician in Laboratory of Prof. Dr. Fyodor Kondrashov — *Oct 2022 - Aug 2023*

- Created *E. coli* mutant libraries of GFPs, sorted them using FACS, improved functional protein prediction using ML
- Devised and implemented a population genetics model for epistasis
- Presented this work as a poster at **SMBE 2025** in Beijing, receiving both the **Young Investigator and Undergraduate Student Travel Awards**

Institute of Science and Technology — *Klosterneuburg, Austria*

Internship in Laboratory of Prof. Dr. Fyodor Kondrashov — *Apr 2022 - Sept 2022*

- Cloned and purified green fluorescent proteins with varying sequence divergence from the wild-type sequence
- Experimentally assessed the physical stability of the proteins and correlated it with the mutational robustness (GitHub)

Bioinformatics institute *Online*

Internship in Laboratory of Prof. Dr. Maria Logacheva *Feb 2022 - May 2022*

- Investigated the expression patterns of egg-cell specific genes in crops

Ruhr University Bochum, Faculty of Biology and Biotechnology *Bochum, Germany*

Internship in Laboratory of Prof. Dr. Ute Krämer *Jan 2021 - Jan 2022*

- Implemented a pipeline to investigate the contribution of CNV to metal tolerance in *A. halleri*

School of Molecular and Theoretical Biology *Barcelona, Spain - Online*

Student in School of Molecular and Theoretical Biology in Laboratory of Prof. Dr. Mikhail Gelfand *Aug 2019*

- Investigated the phylogenetics of intracellular parasites *B. mallei* in Laboratory of Prof. Dr. Mikhail Gelfand in 2019

Teaching and Mentoring Experience

ETH Zürich, Department of Biosystems Science and Engineering *Zürich, Switzerland*

Teaching Assistant in Computational Systems Biology *Oct 2024 - Apr 2025*

- Writing lecture notes for computational systems biology class under Prof. Dr. Jörg Stelling

Okinawa Institute of Science and Technology, Japan *Onna, Japan*

Research and Lab mentor *July 2023 - Aug 2023*

- Guided an intern through the whole process from preparation of *E. coli* GFP library to sorting and preparing the library for sequencing

School of Molecular and Theoretical Biology *Online*

Staff member and Mentor *Aug 2020 - Aug 2022*

- Introduced machine learning, Git, and worked on a research project with high-school students in Laboratory of Prof. Dr. Olga Kalinina in 2022
- Mentored 12 students throughout their stay at the online summer school and beyond, and organized school-wide social activities

Ruhr University Bochum, Faculty of Chemistry and Biochemistry *Bochum, Germany*

Teaching Assistant in Maths under the supervision of Prof. Dr. Lars Schäfer and Dr. Eva Glasmachers *Oct 2020 - Sept 2022*

- Graded assignments and exams for the maths courses (20-50 participants, 500 academic hours) for first-year (bio-)chemistry students

Ruhr University Bochum, Faculty of Chemistry and Biochemistry *Bochum, Germany*

Supervisor in Laboratory Practical under the supervision of Prof. Dr. Rochus Schmid *March 2022*

- Taught basic safety rules in the lab, monitored experiments, and graded lab protocols to 20 students

Skills

Molecular Biology	Cloning; Western Blot; DNA, RNA, and protein extraction; PCR; DNA and protein electrophoresis
Microbiology	*Bacteria*: Transformation; culture; screening; generation of competent cells; genome integration; microscopy; sorting
Programming	Python; R; high-performace computing (SLURM, Unix, Bash); version control (Git, Conda); SQL databases; LaTeX
Bioinformatics	Data analysis and visualization; statistics; NGS analysis workflows (alignment, assembly, etc.); phylogenetics, phylodynamics, microbiomics; theoretical modeling; machine learning and deep learning (PyTorch); scRNA analysis (Scanpy);

Honors

2026	**Recipient,** Pasteur Paris-University (PPU) international doctoral program, 3-year fellowship	*Paris, France*
2025	**Recipient,** Young Investigator and Undergraduate Student Awards for Travelling to SMBE	*Beijing, China*
2023	**Recipient,** ESOP, Excellence Scholarship & Opportunity Program	*Zürich, Switzerland*
2023	**Winner,** Wilke-Award for best Bachelor's degree in Biochemistry	*Bochum, Germany*
2021-2022	**Recipient,** Deutschlandstipendium Scholarship	*Bochum, Germany*
2018, 2019	**Two Bronze Medals,** 29th and 30th International Biology Olympiad (IBO)	*Iran and Hungary*
2018	**Qualification,** Internationale Deutscholympiade 2018 (Goethe-Institut), could not participate due to IBO	*Tbilisi, Georgia*
2018, 2019	**Winner,** National Biology Olympiad	*Tbilisi, Georgia*
2018, 2019	**Winner,** National German Language Olympiad	*Tbilisi, Georgia*

Languages

English	C1-C2, TOEFL score 116 (2025)
German	C1-C2, higher education in German Language and C1 Goethe certificate (very good, 2019)
Georgian	Native language
Russian	Native language
French	A2